Bài tập tính toán

Thứ ngày

$X \in R^{3\times 5}$ giả sử $X = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$ $W_K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ $W_V = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow Q = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}$ $K = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ $V = \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$

$S = \frac{QK^T}{\sqrt{2}} = \frac{\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}}{\sqrt{2}} = \frac{\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}}{\sqrt{2}}$

$= \begin{bmatrix} 0 & 0 & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$

$\text{Softmax}(S_1) = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0{,}4011 & 0{,}4011 & 0{,}1977 \\ 0{,}4011 & 0{,}4011 & 0{,}1977 \end{bmatrix}$

$Z = A \cdot V = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0{,}4011 & 0{,}4011 & 0{,}1977 \\ 0{,}4011 & 0{,}4011 & 0{,}1977 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \\ 0{,}4011 & 0{,}5989 \\ 0{,}4011 & 0{,}5989 \end{bmatrix}$

SON NAM

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10 Bài tập tính positional encoding

Thứ ngày

$d = 5$

$$X = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$PE = \begin{pmatrix} PE(0,0) & PE(0,1) & PE(0,2) & PE(0,3) & PE(0,4) \\ PE(1,0) & PE(1,1) & PE(1,2) & PE(1,3) & PE(1,4) \\ PE(2,0) & PE(2,1) & PE(2,2) & PE(2,3) & PE(2,4) \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 0,8415 & 0,5403 & 0,0251 & 0,9997 & 6,309 \cdot 10^{-4} \\ 0,9093 & -0,4161 & 0,0502 & 0,9997 & 1,2619 \cdot 10^{-3} \end{pmatrix}$$

$$X = X + PE = \begin{bmatrix} 0 & 2 & 0 & 1 & 0 \\ 1,8415 & 0,5403 & 0,0251 & 0,9997 & 6,309 \cdot 10^{-4} \\ 0,9093 & -0,4161 & 0,0502 & 0,9997 & 1,0013 \end{bmatrix}$$

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